ThreatMatrix AI — Team Roles & Contributions

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> **Project:** ThreatMatrix AI — AI-Powered Network Anomaly Detection and Cyber Threat Intelligence System
> **Context:** Bachelor's Degree Senior Project — Computer Science
> **Institution:** ASTU (Adama Science and Technology University)
> **Meeting Use:** Advisor presentation script — read top-to-bottom

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1. Document Header

1.1 Project Identity

ThreatMatrix AI is an enterprise-grade, AI-powered cybersecurity platform that provides real-time network anomaly detection and cyber threat intelligence through a unified War Room command center. The system combines three distinct machine learning models (Isolation Forest, Random Forest, Autoencoder) with multi-provider LLM integration to deliver predictive threat analysis, automated incident response narratives, and actionable intelligence — all presented through a military-grade operational UI inspired by real-world intelligence agency cyber operations centers. *(Source: `MASTER_DOC_PART1_STRATEGY.md` §1.1)*

1.2 Team Roster

Name	Title	One-Line Scope
Kidus Abdula	Lead Senior Software Engineer & Systems Architect	Full architecture, backend, ML pipeline, LLM gateway, capture engine, deployment, technical documentation
Caleb Demelash	Full-Stack Engineer	Next.js 16 frontend, War Room UI, data visualization, API wiring, design system
Dinaol Seyoum	Business Manager	Market research, business plan, budget, presentation materials, advisor communications
Kirubel Tewodros	QA & Tester	Test plans, attack simulation, UAT, Amharic translations, demo rehearsal

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2. Executive Summary of Contributions

2.1 What the Team Built

Over an 8-week development window (February 24 — April 20, 2026), the team shipped ThreatMatrix AI v1.0.0: a production-grade cybersecurity platform running on a live Hostinger KVM VPS, processing real network traffic, with 46 operational REST API endpoints, 10 functional frontend modules, three trained ML models scoring traffic in real-time, an OpenRouter-powered LLM gateway routing across five free-tier models, and a fully bilingual English/Amharic interface.

The system is not a prototype — it is a deployable product. One `docker compose up -d` command launches five containers (PostgreSQL, Redis, FastAPI backend, Scapy capture engine, ML inference worker) behind an Nginx reverse proxy with SSL. The VPS has demonstrated 5+ consecutive days of continuous uptime.

2.2 How Responsibility Was Distributed

Team Member	Hours/Week	Total Hours (8 weeks)	Codebase Share	Deliverable Type
Kidus Abdula (Lead)	40–60	320–480	~60%	Backend, ML, infrastructure, architecture docs
Caleb Demelash (Full-Stack)	30–40	240–320	~30%	Frontend application, UI components, design system
Kirubel Tewodros (Tester)	10–15	80–120	~10% + testing	Test plans, attack scripts, translations, QA
Dinaol Seyoum (Business)	15–20	120–160	Non-code	Business docs, presentations, budget, market research

(Source: `MASTER_DOC_PART1_STRATEGY.md` §1.4; `MASTER_DOC_PART5_TIMELINE.md` §4.2)

The distribution is honest: the Lead Architect's scope dominates the codebase by design, because the project is backend-heavy (ML pipeline, capture engine, LLM gateway, threat intelligence correlation). The Full-Stack Engineer's contribution is concentrated entirely in the Next.js 16 frontend. The QA & Tester's work spans test documentation, attack simulation, and the complete Amharic i18n dictionary. The Business Manager owns every non-technical deliverable required for academic submission and commercial viability.

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3. Feature Ownership Matrix

This matrix reproduces the authoritative allocation from `MASTER_DOC_PART5_TIMELINE.md` §4.1. Each bar (█) represents one unit of ownership: six bars = sole/full ownership, four bars = major ownership, two bars = minor/supporting ownership.

Feature	Kidus Abdula (Lead)	Caleb Demelash (Full-Stack)	Dinaol Seyoum (Business)	Kirubel Tewodros (Tester)
FastAPI Backend	██████			
Database Schema	██████			
Capture Engine	██████			
ML Pipeline	██████			
LLM Gateway	██████			
Alert Engine	██████			
WebSocket Server	██████			
War Room UI	██	██████		

Feature	Kidus Abdula (Lead)	Caleb Demelash (Full-Stack)	Dinaol Seyoum (Business)	Kirubel Tewodros (Tester)
All Other UI		███		
Design System	█	███		
i18n Setup		██		█
Amharic Translations				███
Business Documents			███	
Presentation			███	
Market Research			███	
Test Plan				███
Testing Execution				███
Attack Simulation	██			█
Documentation	██		█	
Deployment	███			

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4. Individual Contribution Profiles

Each profile follows the same five-subsection template for consistent scan-reading during the advisor meeting.

4.1 Kidus Abdula — Lead Senior Software Engineer & Systems Architect

4.1.1 Role Statement

Kidus Abdula served as the Lead Senior Software Engineer and Systems Architect, owning the full technical backbone of ThreatMatrix AI. This included system architecture, backend engineering, machine learning pipeline design, LLM integration, packet capture engine development, deployment infrastructure, and all master technical documentation. The role also encompassed frontend task delegation, architectural review, and final integration unblocking.

4.1.2 Scope of Work

- Authored the complete 5-part Master Documentation (~120 pages) and 2 Progress Reports
- Wrote 23+ daily worklog files in `docs/worklog/` tracking technical progress
- Built the FastAPI backend application factory, 46 REST endpoints, JWT authentication, RBAC (4 roles), and 10 PostgreSQL tables with Alembic migrations
- Developed the Scapy-based capture engine with 5-tuple flow aggregation, 63-feature extraction, and Redis pub/sub publishing

- Implemented the complete ML pipeline: NSL-KDD + CICIDS2017 loaders, Isolation Forest, Random Forest, TensorFlow Autoencoder, ensemble scorer, model manager, training orchestrator, and hyperparameter tuning
- Built the OpenRouter LLM Gateway with multi-model routing, prompt templates, streaming SSE, caching, and budget tracking
- Integrated threat intelligence aggregation (AlienVault OTX, AbuseIPDB, VirusTotal) and the IOC Correlation Engine
- Created the real-time inference worker (Redis subscriber → ensemble scoring → auto-alert creation)
- Implemented ReportLab PDF report generation pipeline
- Deployed the full Docker Compose stack (5 containers) on a Hostinger KVM VPS, configured Nginx reverse proxy, SSL via Let's Encrypt, and production hardening

4.1.3 Concrete Deliverables

Deliverable	Location	Evidence
Master Documentation (5 parts)	`docs/master-documentation/`	`MASTER_DOC_PART1_STRATEGY.md` through `MASTER_DOC_PART5_TIMELINE.md`
Daily Worklogs	`docs/worklog/`	23+ daily log files
ML Evaluation Report	`docs/ThreatMatrix_AI_Day10_Report.md`	Academic metrics and model comparison
FastAPI Backend	`backend/app/`	App factory, routers, services, schemas
Capture Engine	`backend/capture/`	`engine.py`, `flow_aggregator.py`, `feature_extractor.py`
ML Pipeline	`backend/ml/`	`datasets/`, `models/`, `training/`, `inference/`
Ensemble Scorer	`backend/ml/inference/ensemble_scorer.py`	Composite weighted scoring implementation
Inference Worker	`backend/ml/inference/worker.py`	Real-time scoring loop
LLM Gateway	`backend/app/services/llm_gateway.py`	Multi-provider router with fallback chain
Threat Intel Service	`backend/app/services/threat_intel.py`	OTX/AbuseIPDB/VirusTotal aggregation
IOC Correlator	`backend/app/services/ioc_correlator.py`	Flow-to-IOC correlation engine
Reports API	`backend/app/api/v1/reports.py`	ReportLab PDF generation endpoints
Docker Compose Stack	`docker-compose.yml`	5-container orchestration

Deliverable	Location	Evidence
Nginx Configuration	`nginx/nginx.conf`	Reverse proxy + SSL

4.1.4 Quantified Impact

Metric	Value	Source
REST API Endpoints	46 / 46 (100%)	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
ML Models Trained	3 + ensemble	`backend/ml/models/`
Ensemble Accuracy	80.73%	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
Ensemble AUC-ROC	0.9312	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
Average Inference Latency	146 ms	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
IOCs Ingested	1,367	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
PostgreSQL Tables	10 normalized tables	`MASTER_DOC_PART5_TIMELINE.md` §2.1
VPS Uptime	5+ consecutive days	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
Docker Containers	5 healthy containers	`docker-compose.yml`
OpenRouter Models	5 free-tier models routed	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §7
Weekly Effort	40–60 hours/week	`MASTER_DOC_PART5_TIMELINE.md` §4.2
Total Estimated Effort	320–480 hours	`MASTER_DOC_PART5_TIMELINE.md` §4.2
Codebase Ownership	~60%	`MASTER_DOC_PART1_STRATEGY.md` §1.4

4.1.5 Weekly Timeline Cross-Reference

Week	Primary Focus	Key Deliverables
Week 1	Foundation	Monorepo init, Docker Compose, PostgreSQL schema, Alembic, FastAPI skeleton, Redis pub/sub
Week 2	Capture + Core Pipeline	Scapy capture engine, 5-tuple flow aggregation, 63-feature extraction, real-time flow publishing
Week 3	ML Training	NSL-KDD preprocessing, Isolation Forest, Random Forest, Autoencoder training, ensemble scorer

Week	Primary Focus	Key Deliverables
Week 4	Intelligence Integration	LLM Gateway (later migrated to OpenRouter), AI Analyst backend, threat intel aggregator, real-time inference worker, alert engine
Week 5	Feature Depth	Anomaly scoring refinement, PCAP processor, WebSocket broadcasting, CICIDS2017 secondary validation
Week 6	Reports + Enterprise	ReportLab PDF generation, RBAC enforcement, LLM budget tracking, system health monitoring
Week 7	Polish + Optimization	Query tuning, model retraining with optimized hyperparameters, pre-built PCAP demo scenarios, attack simulation scripts
Week 8	Final Push	Production VPS deployment, SSL/Let's Encrypt, security hardening, API documentation finalization

(Source: `MASTER_DOC_PART5_TIMELINE.md` §3)

4.2 Caleb Demelash — Full-Stack Engineer

4.2.1 Role Statement

Caleb Demelash served as the Full-Stack Engineer, owning the entire Next.js 16 frontend application, the Deep Space design system, and all user-facing visualization components. The role required translating backend APIs into a military-grade War Room interface, implementing real-time WebSocket consumption, building 10 distinct module pages, and ensuring responsive, animated, accessible UI across the application.

4.2.2 Scope of Work

- Initialized and scaffolded the Next.js 16 App Router project with TypeScript strict mode
- Implemented the Deep Space dark-theme design system using pure CSS variables (no Tailwind), including glassmorphism panels, JetBrains Mono data typography, Inter UI typography, and cyber-cyan accent colors
- Built the War Room command center with 9 live components: MetricCard, ThreatMap (Deck.gl + Maplibre), ProtocolChart, TrafficTimeline, TopTalkers, LiveAlertFeed, ThreatLevel gauge, AIBriefingWidget, and GeoDistribution
- Developed the AI Analyst chat interface with streaming SSE responses and the AnalysisPanel/QueryBuilder
- Built the Alert Console with severity filtering, sortable data tables, and detail drawers
- Created the ML Operations dashboard with model comparison tables, confusion matrices, ROC curves, feature importance charts, and retrain polling
- Implemented the Intel Hub, Forensics Lab, Reports, Threat Hunt, Administration, and Network Flow module UIs
- Authored custom React hooks for data fetching and real-time updates: `useWebSocket`, `useFlows`, `useAlerts`, `useLLM`
- Built the API client layer (`api.ts`) and WebSocket client (`websocket.ts`)
- Added responsive layouts, Framer Motion page transitions, and polished loading/empty/error states
- Participated in the Day 18 integration sprint wiring remaining frontend pages to live VPS APIs

4.2.3 Concrete Deliverables

Deliverable	Location	Evidence
Next.js 16 App Shell	`frontend/app/`	10+ module pages with App Router
Design System	`frontend/app/globals.css`	CSS variables, glassmorphism, color system
Shared Components	`frontend/components/shared/`	`GlassPanel.tsx`, `DataTable.tsx`, `StatusBadge.tsx`, `LoadingState.tsx`
War Room Components	`frontend/components/war-room/`	`ThreatMap.tsx`, `ThreatGauge.tsx`, `LiveAlertFeed.tsx`, `MetricCard.tsx`, `ProtocolChart.tsx`, `TrafficTimeline.tsx`, `TopTalkers.tsx`, `GeoDistribution.tsx`, `AIBriefingWidget.tsx`
AI Analyst Components	`frontend/components/ai-analyst/`	`ChatInterface.tsx`, `QuickActions.tsx`, `ContextPanel.tsx`
Alert Components	`frontend/components/alerts/`	`AlertTable.tsx`, `AlertDetail.tsx`, `AlertTimeline.tsx`
ML Ops Dashboard	`frontend/app/ml-ops/`	Model comparison, confusion matrices, ROC curves, feature importance
Custom Hooks	`frontend/hooks/`	`useWebSocket.ts`, `useFlows.ts`, `useAlerts.ts`, `useMLModels.ts`, `useLLM.ts`
API Client	`frontend/lib/api.ts`	Fetch wrapper with auth headers
WebSocket Client	`frontend/lib/websocket.ts`	Real-time connection manager
Administration UI	`frontend/app/admin/`	User management, config, LLM budget, audit pages

4.2.4 Quantified Impact

Metric	Value	Source
Frontend Modules Shipped	10 / 10	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §5
War Room Components	9 live widgets	`frontend/components/war-room/`
Frontend Pages	10+ routed pages	`frontend/app/`
Custom React Hooks	5 data hooks	`frontend/hooks/`
Shared UI Components	4+ reusable primitives	`frontend/components/shared/`
Zero Tailwind CSS	Pure CSS variables	`frontend/app/globals.css`
Weekly Effort	30–40 hours/week	`MASTER_DOC_PART5_TIMELINE.md` §4.2

Metric	Value	Source
Total Estimated Effort	240–320 hours	`MASTER_DOC_PART5_TIMELINE.md` §4.2
Codebase Ownership	~30%	`MASTER_DOC_PART1_STRATEGY.md` §1.4

4.2.5 Weekly Timeline Cross-Reference

Week	Primary Focus	Key Deliverables
Week 1	UI Foundation	Next.js 16 init, layout, sidebar, dark theme shell, CSS variables, `GlassPanel`
Week 2	Core Components	War Room grid layout, `MetricCard`, `StatusBadge`, `DataTable`, `useWebSocket.ts`
Week 3	Maps & Charts	`ThreatMap` (Deck.gl + Maplibre), `TrafficTimeline`, `ProtocolChart`, Network Flow layout
Week 4	Intelligence UI	AI Analyst chat interface (streaming), Intel Hub, Alert Console table + drawer
Week 5	Feature Depth	`LiveAlertFeed`, `TopTalkers`, `GeoDistribution`, Forensics Lab UI, ML Ops dashboard
Week 6	Enterprise UI	Reports module, Admin panel, LLM Budget dashboard, enhanced Network Flow
Week 7	Polish + i18n	Responsive layouts, Framer Motion animations, loading/error states, i18n scaffold
Week 8	Final Fixes	UI bug fixes, pixel-perfect War Room polish, theme toggle (stretch)

(Source: `MASTER_DOC_PART5_TIMELINE.md` §3)

4.3 Dinaol Seyoum — Business Manager

4.3.1 Role Statement

Dinaol Seyoum served as the Business Manager, owning all non-technical deliverables required for academic submission, commercial credibility, and advisor communication. The role encompassed market analysis, business planning, financial modeling, presentation materials, legal documentation, and end-user documentation. Dinaol's work provided the commercial and academic context that frames ThreatMatrix AI as a viable enterprise product, not merely a student project.

4.3.2 Scope of Work

- Conducted competitive market analysis: competitor matrix (Splunk, QRadar, Elastic SIEM, Wazuh, Snort), SWOT analysis, and pricing benchmarks feeding `MASTER_DOC_PART1_STRATEGY.md` §5

- Authored the Business Plan & Revenue Model with tiered SaaS pricing (Sentinel / Guardian / Warden), unit economics, CAC/MRR/margin tables, and revenue projections for Years 1–3
- Defined target customer personas: government (INSA), financial (CBE, Dashen, Awash, Telebirr), telecom (Ethio Telecom, Safaricom Ethiopia), universities, and SMEs
- Created the `Budget_Allocation.md` / `.pdf` document tracking project finances
- Drafted the legal framework document authorizing VPS traffic capture
- Collaborated on the Defense Presentation script (`docs/DEFENSE_PRESENTATION_SCRIPT.md`)
- Authored advisor communications including the reschedule request document
- Produced the end-user manual PDF (Week 8 deliverable)
- Delivered presentation slides and demo script contributions
- Presented the Problem/Market section (Section 2) and Business Case section (Section 8) during demo day per `MASTER_DOC_PART5_TIMELINE.md` §8.1

4.3.3 Concrete Deliverables

Deliverable	Location	Evidence
Budget Allocation	`docs/Budget_Allocation.md` / `.pdf`	$170 total budget, $3.40 spent, $166.60 remaining
Defense Presentation Script	`docs/DEFENSE_PRESENTATION_SCRIPT.md`	Collaborative business/presentation artifact
Advisor Reschedule Request	`docs/ThreatMatrix_AI_Defense_Reschedule_Request.md`	Formal advisor-comms artifact
Market & Competitive Analysis	Embedded in `MASTER_DOC_PART1_STRATEGY.md` §5	Competitor matrix, SWOT, pricing benchmarks
Business Plan & Revenue Model	Embedded in `MASTER_DOC_PART1_STRATEGY.md` §6	Tiered SaaS, unit economics, projections
Target Customer Personas	Embedded in `MASTER_DOC_PART1_STRATEGY.md` §7	5 segments with budget ranges and decision cycles
User Manual	Week 8 PDF deliverable	End-user operating guide
Presentation Slides	Week 8 deliverable	Advisor meeting and defense slides

4.3.4 Quantified Impact

Metric	Value	Source
Budget Tracked	$170 total / $3.40 spent / $166.60 remaining	`docs/Budget_Allocation.md`
Revenue Tiers Defined	3 (Sentinel / Guardian / Warden)	`MASTER_DOC_PART1_STRATEGY.md` §6.1
Customer Segments Mapped	5 primary segments	`MASTER_DOC_PART1_STRATEGY.md` §7
Gross Margin Modeled	85–92%	`MASTER_DOC_PART1_STRATEGY.md` §6.3
Demo Sections Owned	2 of 9 (Problem/Market + Business Case)	`MASTER_DOC_PART5_TIMELINE.md` §8.1
Weekly Effort	15–20 hours/week	`MASTER_DOC_PART5_TIMELINE.md` §4.2
Total Estimated Effort	120–160 hours	`MASTER_DOC_PART5_TIMELINE.md` §4.2
Codebase Ownership	Non-code deliverables	`MASTER_DOC_PART1_STRATEGY.md` §1.4

4.3.5 Weekly Timeline Cross-Reference

Week	Primary Focus	Key Deliverables
Week 1	Legal & Framework	VPS traffic-capture authorization document
Week 2	Market Research	Competitive analysis report
Week 3	Competitor Deep-Dive	Completed competitor analysis, user personas
Week 4	Use Cases & Personas	User personas, use-case documentation
Week 5	Presentation Draft	Demo script draft, slide deck start
Week 6	Financial Modeling	Revenue projection model, business plan polish
Week 7	Final Business Docs	Final business plan, presentation slides
Week 8	Submission Package	User manual PDF, presentation rehearsal

(Source: `MASTER_DOC_PART5_TIMELINE.md` §3)

4.4 Kirubel Tewodros — QA & Tester

4.4.1 Role Statement

Kirubel Tewodros served as the QA & Tester, owning the quality assurance lifecycle, attack simulation infrastructure, dataset validation, API testing, integration testing, and the complete Amharic localization of the frontend. The role ensured that every component — from ML model accuracy to end-to-end Docker Compose flows — was verified before demo day, and that the system was fully accessible to Amharic-speaking analysts.

4.4.2 Scope of Work

- Authored the comprehensive test plan covering unit tests, API tests, ML accuracy tests, integration tests, UAT, and performance tests (`MASTER_DOC_PART5_TIMELINE.md` §7.1)
- Created attack simulation scripts: nmap port-scan, hping3 DDoS flood, iodine DNS-tunneling, hydra SSH brute-force, plus benign baseline traffic — each mapped to expected severity levels (`MASTER_DOC_PART5_TIMELINE.md` §7.3)
- Validated dataset integrity for NSL-KDD (125,973 train / 22,544 test samples) and the CICIDS2017 loader
- Executed API testing across all 46 REST endpoints using Postman and curl; recorded results in test reports
- Performed integration and end-to-end testing of the full capture → Redis → ML scoring → alert → WebSocket → UI pipeline across the Docker Compose stack
- Authored the complete Amharic i18n dictionary in `frontend/messages/am.json`, including translations for War Room, Alerts, AI Analyst, and admin surfaces
- Conducted UAT execution: demo scenario dry-runs and go/no-go checklist sign-off
- Drove the 20-minute demo walkthrough rehearsals in Week 8
- Validated the pre-demo checklist: VPS uptime, LLM availability, IOC freshness, PDF report generation, Amharic locale load, backup video readiness, second laptop configuration

4.4.3 Concrete Deliverables

Deliverable	Location	Evidence
Test Plan Document	`MASTER_DOC_PART5_TIMELINE.md` §7.1	Layered test strategy (unit/API/ML/integration/UAT/performance)
Attack Simulation Scripts	`MASTER_DOC_PART5_TIMELINE.md` §7.3	nmap, hping3, iodine, hydra scenarios with severity mapping
API Test Results	Test report documents	Postman/curl coverage of 46 endpoints
Integration Test Report	End-to-end validation	Full Docker Compose pipeline verification
Amharic Translations	`frontend/messages/am.json`	Complete i18n dictionary
UAT Report	Week 8 deliverable	Go/no-go checklist, demo scenario validation
Pre-Demo Checklist Validation	Week 8 execution	VPS uptime, LLM health, IOC sync, PDF generation, Amharic load

4.4.4 Quantified Impact

Metric	Value	Source
Test Layers Owned	6+ (unit, API, ML, integration, UAT, performance)	`MASTER_DOC_PART5_TIMELINE.md` §7.1
Attack Scenarios Scripted	5 (port scan, DDoS, DNS tunnel, SSH brute-force, benign baseline)	`MASTER_DOC_PART5_TIMELINE.md` §7.3
API Endpoints Tested	46 / 46	Integration test coverage
Dataset Samples Verified	NSL-KDD 125,973 train / 22,544 test	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §6
Languages Supported	2 (English + Amharic)	`frontend/messages/am.json`
Demo Rehearsals Driven	20-minute walkthroughs in Week 8	`MASTER_DOC_PART5_TIMELINE.md` §8.3
Weekly Effort	10–15 hours/week	`MASTER_DOC_PART5_TIMELINE.md` §4.2
Total Estimated Effort	80–120 hours	`MASTER_DOC_PART5_TIMELINE.md` §4.2
Codebase Ownership	~10% + testing scope	`MASTER_DOC_PART1_STRATEGY.md` §1.4

4.4.5 Weekly Timeline Cross-Reference

Week	Primary Focus	Key Deliverables
Week 1	Test Planning	Test plan document draft, dev environment setup
Week 2	Test Data Generation	Sample flow/alert data scripts
Week 3	Dataset Validation	NSL-KDD and CICIDS2017 data integrity verification
Week 4	API Testing	Postman/curl test suite for all endpoints
Week 5	Integration Testing	End-to-end capture → alert → UI test results
Week 6	E2E Testing	Full system test report
Week 7	i18n + UAT	Amharic translation review, UAT execution
Week 8	Final QA + Rehearsal	Final QA pass, demo day rehearsal, go/no-go decision

(Source: `MASTER_DOC_PART5_TIMELINE.md` §3)

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5. Shared Responsibilities & Collaboration Points

No team member worked in complete isolation. The following collaborative interfaces were critical to shipping v1.0:

Collaboration Area	Primary Owner	Supporting Owner	Description
i18n Setup	Caleb Demelash	Kirubel Tewodros	Caleb scaffolded the Next.js i18n framework and `messages/` directory structure; Kirubel authored 100% of the Amharic content in `frontend/messages/am.json`
Attack Simulation	Kidus Abdula	Kirubel Tewodros	Kidus built the capture infrastructure and VPS environment that enabled live attack detection; Kirubel authored and executed the attack scripts (nmap, hping3, iodine, hydra)
Documentation	Kidus Abdula	Dinaol Seyoum	Kidus authored all technical and architectural documentation (Master Docs, worklogs, ML reports); Dinaol authored all business-facing documentation (business plan, user manual, budget, presentation materials)
Demo Day Presentation	All members	—	Kidus presented the Hook, Architecture, Live Demo, ML Results, and Close; Caleb presented the Enterprise section (Admin/Reports); Dinaol presented Problem/Market and Business Case; Kirubel managed backstage QA and backup execution

6. Timeline Alignment (Weeks 1–8)

The table below reproduces the per-week owner assignments from `MASTER_DOC_PART5_TIMELINE.md` §3, showing how each member's scope interlocked across the 8-week lifecycle.

Week	Theme	Kidus Abdula	Caleb Demelash	Dinaol Seyoum	Kirubel Tewodros
1	Foundation	Monorepo, Docker Compose, PostgreSQL schema, Alembic, FastAPI skeleton, Redis	Next.js 16 init, layout, sidebar, dark theme, CSS variables, `GlassPanel`	Legal framework for VPS capture	Test plan draft, dev env setup
2	Capture + Core UI	Scapy capture engine, 5-tuple flow aggregation, 63-feature extraction, Redis pub/sub	War Room grid, `MetricCard`, `StatusBadge`, `DataTable`, `useWebSocket.ts`	Market research document	Test data generation scripts

Week	Theme	Kidus Abdula	Caleb Demelash	Dinaol Seyoum	Kirubel Tewodros
3	ML Pipeline	NSL-KDD preprocessing, 3 ML models trained, ensemble scorer, evaluation framework	`ThreatMap` (Deck.gl + Maplibre), `TrafficTimeline`, `ProtocolChart`, Network Flow layout	Competitor analysis report	Dataset validation testing
4	Intelligence Integration	LLM Gateway, AI Analyst backend, threat intel aggregator, inference worker, alert engine	AI Analyst chat UI (streaming), Intel Hub, Alert Console table + drawer	User personas, use-case docs	API testing (Postman/curl)
5	Feature Depth	Anomaly scoring refinement, PCAP processor, WebSocket broadcasting, CICIDS2017 validation	`LiveAlertFeed`, `TopTalkers`, `GeoDistribution`, Forensics Lab UI, ML Ops dashboard	Demo script draft, slide deck start	Integration testing
6	Reports + Enterprise	ReportLab PDF generation, RBAC enforcement, LLM budget tracking, system health	Reports module UI, Admin panel, LLM Budget dashboard, enhanced Network Flow	Revenue projection model	End-to-end testing
7	Polish + i18n	Performance optimization, model retraining, pre-built PCAP demo scenarios, attack scripts	Responsive design, Framer Motion animations, loading/error states, i18n scaffold	Final business plan, presentation slides	UAT testing, Amharic translation review
8	Final Push	Production deployment, SSL, security hardening, API docs finalization	Final UI bug fixes, pixel-perfect War Room polish	Presentation rehearsal, user manual PDF	Final QA, demo day rehearsal, go/no-go

7. Version Tag Ownership (v0.1.0 → v1.0.0)

Each version milestone below cross-references `MASTER_DOC_PART5_TIMELINE.md` §5.1 and attributes the primary drivers.

Version	Date	Milestone	Primary Driver(s)
v0.1.0	Week 1	Project skeleton, database, auth, UI shell	Kidus Abdula
v0.2.0	Week 2	Capture engine, flow storage, War Room layout	Kidus Abdula
v0.3.0	Week 3	ML models trained, basic scoring, map + charts	Kidus Abdula
v0.4.0	Week 4	LLM integration, AI Analyst, threat intel, alerts	Kidus Abdula + Caleb Demelash
v0.5.0	Week 5	PCAP forensics, ML dashboards, full War Room	Caleb Demelash + Kidus Abdula; validated by Kirubel Tewodros
v0.6.0	Week 6	Reports, admin, RBAC, budget tracking	Kidus Abdula (backend) + Caleb Demelash (admin UI) + Dinaol Seyoum (business polish)
v0.7.0	Week 7	Polish, animations, i18n, demo scenarios	Caleb Demelash + Kirubel Tewodros + Dinaol Seyoum
v1.0.0	Week 8	Production deployment, final fixes, documentation	Kidus Abdula (deployment lead); whole team (rehearsal and sign-off)

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8. Closing — Combined Team Outcome

ThreatMatrix AI v1.0.0 is the sum of four distinct, necessary contributions. The Lead Architect's scope dominated the codebase by design — the project is fundamentally a backend-and-ML-heavy system — but v1.0 does not ship without every member executing their charter.

8.1 v1.0 System Metrics

Metric	Value	Source
API Endpoints Live	46 / 46	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
Frontend Modules Shipped	10 / 10	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §5
ML Models + Ensemble	3 models + ensemble scorer	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §6
Ensemble Accuracy	80.73%	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
Ensemble AUC-ROC	0.9312	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
Average Inference Latency	146 ms	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4

Metric	Value	Source
IOCs Ingested	1,367	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
VPS Continuous Uptime	5+ days	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §4
Budget Spent / Total	$3.40 / $170	`docs/Budget_Allocation.md`
LLM Operational Cost	$0.00 / month (OpenRouter free tier)	`PRESENTATION_SCRIPT_ADVISOR_MEETING.md` §7
UI Languages	English + Amharic	`frontend/messages/am.json`
Docker Containers	5 healthy containers	`docker-compose.yml`
PostgreSQL Tables	10 normalized tables	`MASTER_DOC_PART5_TIMELINE.md` §2.1

8.2 Closing Statement

Each contribution was necessary. The Lead Engineer authored the architectural backbone that makes the system technically credible. The Full-Stack Engineer built the visual interface that makes the system demoable and sellable. the QA & Tester ensured the system was verified, localized, and rehearsal-ready. The Business Manager provided the commercial and academic framing that elevates the project from a class assignment to a viable product. v1.0 required all four scopes; no single member could have delivered it alone.

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9. Appendix — Evidence Index

This index maps claims in the present document to their source artifacts.

9.1 Technical Artifacts

Claim	Artifact Path
46 REST API endpoints	`backend/app/api/v1/`
FastAPI application factory	`backend/app/main.py`
JWT auth + RBAC	`backend/app/dependencies.py`, `backend/app/services/auth_service.py`
Capture engine (Scapy)	`backend/capture/engine.py`
Flow aggregator	`backend/capture/flow_aggregator.py`
63-feature extractor	`backend/capture/feature_extractor.py`
NSL-KDD loader	`backend/ml/datasets/nsl_kdd.py`
CICIDS2017 loader	`backend/ml/datasets/cicids2017.py`
Isolation Forest model	`backend/ml/models/isolation_forest.py`
Random Forest model	`backend/ml/models/random_forest.py`

Claim	Artifact Path
Autoencoder model	`backend/ml/models/autoencoder.py`
Ensemble scorer	`backend/ml/inference/ensemble_scorer.py`
Inference worker	`backend/ml/inference/worker.py`
LLM Gateway	`backend/app/services/llm_gateway.py`
Threat intel service	`backend/app/services/threat_intel.py`
IOC Correlator	`backend/app/services/ioc_correlator.py`
Report generation	`backend/app/api/v1/reports.py`, `backend/app/services/report_generator.py`
Alembic migrations	`backend/alembic/versions/`
Docker Compose stack	`docker-compose.yml`
Nginx reverse proxy	`nginx/nginx.conf`

9.2 Frontend Artifacts

Claim	Artifact Path
Next.js 16 app shell	`frontend/app/`
Design system (CSS variables)	`frontend/app/globals.css`
GlassPanel component	`frontend/components/shared/GlassPanel.tsx`
War Room components	`frontend/components/war-room/`
AI Analyst components	`frontend/components/ai-analyst/`
Alert components	`frontend/components/alerts/`
Custom hooks	`frontend/hooks/`
API client	`frontend/lib/api.ts`
WebSocket client	`frontend/lib/websocket.ts`
English translations	`frontend/messages/en.json`
Amharic translations	`frontend/messages/am.json`

9.3 Documentation & Business Artifacts

Claim	Artifact Path
5-part Master Documentation	`docs/master-documentation/MASTER_DOC_PART1_STRATEGY.md` through `MASTER_DOC_PART5_TIMELINE.md`
Daily worklogs	`docs/worklog/`

Claim	Artifact Path
ML Evaluation Report	`docs/ThreatMatrix_AI_Day10_Report.md`
Advisor Meeting Script	`docs/PRESENTATION_SCRIPT_ADVISOR_MEETING.md`
Defense Presentation Script	`docs/DEFENSE_PRESENTATION_SCRIPT.md`
Budget Allocation	`docs/Budget_Allocation.md` / `.pdf`
Advisor Reschedule Request	`docs/ThreatMatrix_AI_Defense_Reschedule_Request.md`

End of Document — ThreatMatrix AI Team Roles & Contributions v1.0.0